

FINAL GITHUB & ENGINEERING WORKFLOW SPECIFICATION

Project ID: SIH26034
Product: NyayaDrishti-LM
Governing Rule: Simplicity • Determinism • Zero Drift
Status: FROZEN

1. Repository Structure

sih26034-nyayadrishti/	
├ .github/workflows/	# CI pipeline (lint, unit tests, schema validation)
├ docs/	# Frozen architectural and requirements
specifications	
├ data/	
│ └ synthetic/	# DS-SYNTH-001 procedural vector label generator &
tests	
│ └ pilot_50_skus/	# Calibrated physical retail packages with vernier
caliper ground truth	
│ └ public_benchmarks/	# Filtered Indic scene text evaluation splits
(BSTD, IndicSTR12)	
├ models/	
│ └ onnx/	# INT8 quantized CPU models (DBNet++, PP-OCrv4)
│ └ configs/	# Hyperparameters, anchors, confidence thresholds
├ core/	
│ └ quality_gate/	# Blur (Laplacian) and specular glare (HSV) filters
(Member 1)	
│ └ homography/	# ArUco and ISO 7810 planar rectification & scale S
(Member 1)	
│ └ ocr_engine/	# Multilingual DBNet++ + PP-OCrv4 + Tesseract
pipeline (Member 2)	
│ └ extractor/	# Deterministic regex, spatial proximity graph,
address NER (Member 3)	
│ └ rule_engine/	# Declarative AST evaluator & temporal epoch router
(Member 4)	
│ └ evidence/	# SHA-256 Merkle DAG & Section 63 BSA PDF dossier
engine (Member 5)	
├ client/	# Cross-platform UI (React + Vite + Tailwind /
Electron) (Member 6)	
│ └ src/components/	# Viewfinder HUD, Multi-Panel Canvas, Review
Interface	
│ └ src/services/	# Local API client, offline SQLite storage adapter
├ server/	# Lightweight local FastAPI / Express offline
backend	
└ tests/	# PyTest regression suites, golden run fixtures,

2. Branching & Merging Strategy (Strict Trunk-Based)

- **main**: Production-ready, demo-safe code. Direct pushes are strictly **prohibited**.
 - **dev**: Integration branch. Automated CI must pass before merging into **dev**.
 - Feature Branches: Named strictly as: **feat/m<member_id>-<subsystem>** (e.g., **feat/m1-homography**, **feat/m4-ast-rules**).
 - Hotfix Branches: **hotfix/<issue-name>**.
-

3. Pull Request (PR) Policy & Code Review Rules

1. Every PR must solve a specific MRS requirement ID (e.g., **MRS-03: Implement ArUco Planar Homography**).
 2. Every PR must pass all local unit tests before submission.
 3. Every PR must be reviewed and approved by at least **one other team member** whose interface is affected.
 4. **No AGPL-3.0 Dependencies**: Any PR introducing an AGPL-licensed package (e.g., **ultralytics**) will be rejected automatically by CI.
 5. PR merges must be squashed and rebased to maintain a clean, linear Git history.
-

4. Definition of Done (DoD)

A feature is considered **DONE** if and only if:

1. It strictly conforms to the JSON schema defined in **07_API_AND_INTERFACE_CONTRACTS.md**.
2. It has passing unit tests with **> 85%** code coverage for mathematical and rule logic.

3. It executes locally on CPU within the latency budget specified in

[03_FINAL_ARCHITECTURE.md](#).

4. It handles failure gracefully without throwing unhandled exceptions to the UI.

5. It is documented in the central integration test suite.